Omikumar Bhavinkumar Makadia

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Education

Rutgers University **Aug 2024 – ongoing**
PhD in Electrical and Computer Engineering *GPA: 3.75/4.00*
Relevant Coursework: Probability and Statistical Inference for Data Science, Introduction to Data Structures and Algorithms, Regression & Time Series Analysis, Probability, Signal Processing and ML in High Dimensions, Communication networks, Digital Signal Analytics, Optimization for Machine Learning, Inference and Machine Learning for Engineers

Ahmedabad University **Aug 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering

Experience

Graduate Research Assistant **Jan 2025 - ongoing**
Wireless Information Network Laboratory *New Brunswick, USA*
- Actively working on Artificial Intelligence/Machine learning for wireless systems and evaluation, at the Wireless Information Network LABoratory (WINLAB).
- Currently working on AI/ML-based direction of arrival and interference nulling solution for multiple link use cases involving reflective intelligent surfaces or tags, supported by funding from the National Science Foundation (NSF), and being part of two contributions to the top-tier conferences.

Undergraduate Researcher **Feb 2022 – May 2024**
Machine Intelligence, Computing and xG Networks Laboratory *Ahmedabad, India*
- Conducted independent Contributive research at the Machine Intelligence, Computing, and xG Networks (MICxN) Lab, deriving two publications in collaboration with SVNIT and IIT Bombay.

Projects

Generalization in One-Step contextual bandit based DoA Estimation with Passive Backscatter Tags **Jan 2026**
- Secured acceptance at the 51st IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), designing and implementing a data-driven Direction-of-Arrival (DoA) estimation framework using supervised DNNs and one-step contextual bandit-based reinforcement learning.
- Evaluated model generalization under severe train-test distribution shifts, including low SNR regimes, varying Rician K-factors, tag reflection coefficients, and synchronization mismatches.
- Conducted simulation-based validation across diverse tag topologies, including uniform linear arrays and randomized geometries, highlighting robustness to topology and geometry variations.

DoA of Multiple Sources with an Array of Passive Tags Instead of an Antenna Array | *Matlab* **Oct 2025**
- Accepted at the 43rd Annual IEEE MILlitary COMmunications conference (MILCOM) advancing passive DoA estimation research.
- Collaborated on the derivation of Cramér–Rao Bounds and Signal processing for multi-source Direction of Arrival estimation and passive signal processing techniques to strengthen Spectrum Situational Awareness.

Enhancing 6G mmWave Beam Prediction in V2I With Class Imbalance Mitigation | *GAN* **Aug 2024**
- Independently presented an innovative beam prediction approach at the 35th International Symposium on Personal, Indoor, and Mobile Radio Communications (PIMRC), leveraging Tabular GAN to tackle low sample size and class imbalance, achieving a 26% reduction in power loss and a 6% improvement in top-1 beam accuracy compared to traditional methods.

Vehicle-to-Infrastructure Location-Aided Beam Forecasting in 6G | *Random Forest, LSTM* **Jan 2024**
- Presented a peer-reviewed paper at the 16th International Conference on COMmunication Systems & NETworkS (COMSNETS), introducing and validating a novel methodology that slashed training overhead by 86% while achieving 99% reliability in Connected Vehicles and Autonomous Driving, and substantially reduced computational demands by minimizing the number of beams requiring training.

Technical Skills

Computing and Programming: Python, MATLAB, Data Structures and Algorithms
Machine Learning, AI MLOps/AI Lifecycle (Data ingestion, feature engineering, training, inference, evaluation), Foundation Models (LLMs, RAG, agentic AI), Supervised/Unsupervised Learning, Deep Learning, Reinforcement Learning (PPO, TRPO, A2C), Representation Learning, Domain Adaptation, Model Generalization, Imbalanced Learning, Self-Supervised Learning, PyTorch, Scikit-learn, Tensorflow
Signal Processing and Wireless Systems: Direction-of-Arrival (DoA) Estimation, Channel Modeling, Cramér–Rao Bounds (CRB), Array Signal Processing, Passive Backscatter Communication, Spectrum Situational Awareness, mmWave Communications, Interference nulling